D 41988 (Pages : 3) Name.....

Reg. No.....

FOURTH SEMESTER B.Sc. DEGREE EXAMINATION, APRIL 2018

(CUCBCSS—UG)

Complementary Course

MAT 4C 04—MATHEMATICS

Time : Three Hours Maximum : 80 Marks

Part A (Objective Type)

Answer **all** *questions.*
Each question carries 1 *mark.*

1. What do you mean by fundamental period ?
2. Write the characteristic equation of the differential equation $2y'' + 10y' + 25y = 0$.
3. Write the 2 dimensional wave equation.
4. Give an example of a function which is neither odd nor even.
5. State the existence and uniqueness theorem for initial value problem.
6. Find the Laplace transform of $f''(t)$.
7. Find the Wronskian of $e^{\lambda x}$ and $xe^{\lambda x}$.
8. Find $L^{-1}\left(\frac{1}{s/a - 1}\right)$.
9. Find L $(t + e^t)$.
10. State the second shifting theorem of Laplace transform.
11. Write the error estimate for Simpson's rule.
12. Write the formula for Runge-Kutta method.

(12 × 1 = 12 marks)

Part B (Short Answer Type)

Answer any **nine** *questions.*
Each question carries 2 *marks.*

13. Find the Fourier cosine series of $f(x) = \pi - x, 0 < x < \pi$.
14. Show that $u = e^{-w^2c^2t} \sin wx$ is a solution of heat equation.
15. Apply Picard's iteration to solve $y' = y - x^2$, $y(0) = 1$. Also find y (0, 1) and y (0, 2).
16. By reducing to first order solve $y'' + (y')^2 = 0$.

Turn over

17. Find fundamental set of solutions of $2t^2y'' + 3ty' - y = 0 \quad t > 0$. Given that $y_1(t) = t^{-1}$ is a solution.

18. Solve $y'' + 8y' + 16y = 0$.

19. Find the inverse transform of $\frac{4}{(s+1)(s+2)}$.

20. Solve the initial value problem $y'' + y' - 6y = 1 \quad y(0) = 0, \quad y'(0) = 1$.

21. Find $\mathrm{L}\left(te^{-t}\sin t\right)$.

22. Using convolution property find $\mathrm{L}^{-1}\left[\frac{1}{s\left(s^2+4^2\right)}\right]$.

23. Use Trapezoidal rule with $n = 4$ to estimate $\int_1^2 x^2 dx$.

24. Solve the partial differential equation $u_{yy} + 4u = 0$.

(9 × 2 = 18 marks)

Part C (Short Essays)

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. Solve $\left(x^2\mathrm{D}^2 + 3x\mathrm{D} + 1\right)y = 0$.

26. Find the general solution of the differential equation $y'' - 2y' + 5y = 5x^3 - 6x^2 + 6x$.

27. Find the Laplace transform of $e^{-2t}u(t-3)$.

28. Find $\mathrm{L}^{-1}\left(\frac{se^{-2s}}{s^2+\pi^2}\right)$.

29. Find the Fourier series expansion of :

$$f(x) = \begin{cases} \sin x & 0 < x < \pi \\ 0 & \pi < x < 2\pi \end{cases}.$$

30. Solve $xy'' - y' = (3+x)x^2e^x$.

31. Find the solution of $u_x + u_y = 0$ by separating variables.

32. Find approximate solution of $y' + y = e^x \quad y(0) = 0$.

33. Obtain the half-range cosine series of $f(x) = x$ when $0 < x < 2$.

(6 × 5 = 30 marks)

Part D

Answer any **two** *questions.*
Each question carries 10 *marks.*

34. Solve the non-homogeneous equation : $y'' - y' - 2y = 10\cos x$.

35. Solve the initial value problem $y'' + 2y' + 2y = r(t)$

where $r(t) = \begin{cases} 10 \sin 2t & \text{if } 0 < t < \pi \\ 0 & \text{if } t > \pi \end{cases}$

$y(0) = 1 \quad y'(0) = -5.$

Using Laplace transforms.

36. Find the Fourier series of the functions

$$f(x) = \begin{cases} x + x^2 & -\lambda < x < \pi \\ \pi^2 & x = \pm \pi \end{cases}$$

Deduce that $1 + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\ldots = \frac{\pi^2}{6}$.

(2 × 10 = 20 marks)